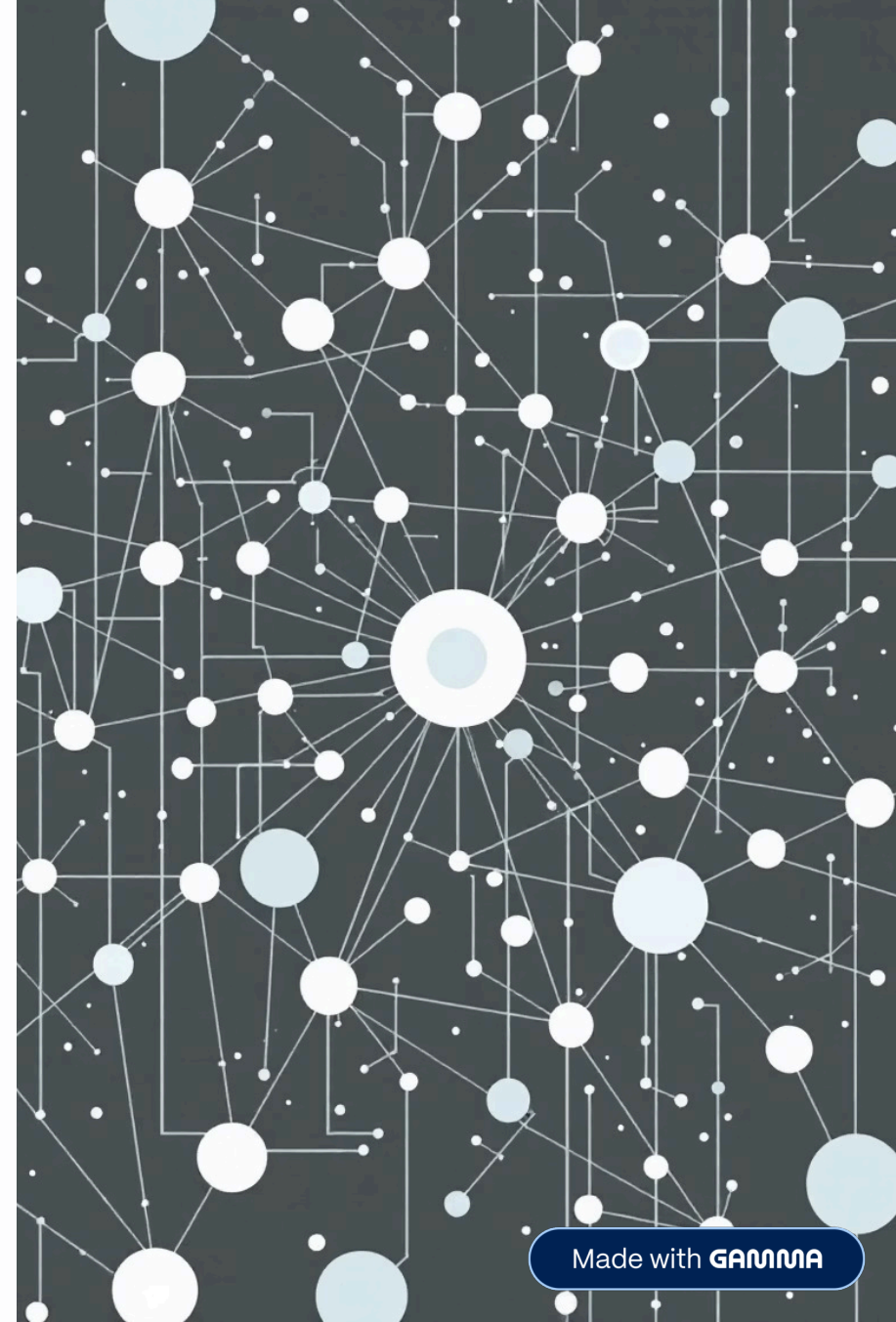


Knowledge Graphs: Structuring Information for Advanced NLP



Knowledge Graphs

A **Knowledge Graph (KG)** is a structured representation of knowledge that captures entities and their relationships. In the context of Natural Language Processing (NLP), KGs provide a contextual understanding of information extracted from text, moving beyond simple keyword matching to nuanced semantic comprehension.

Addressing Information Overload

The exponential growth of heterogeneous, unstructured data necessitates efficient methods for integration and analysis.

Enhancing Semantic Understanding

Unlike traditional statistical methods, KGs utilize semantics to allow for more accurate data extraction and sophisticated reasoning.

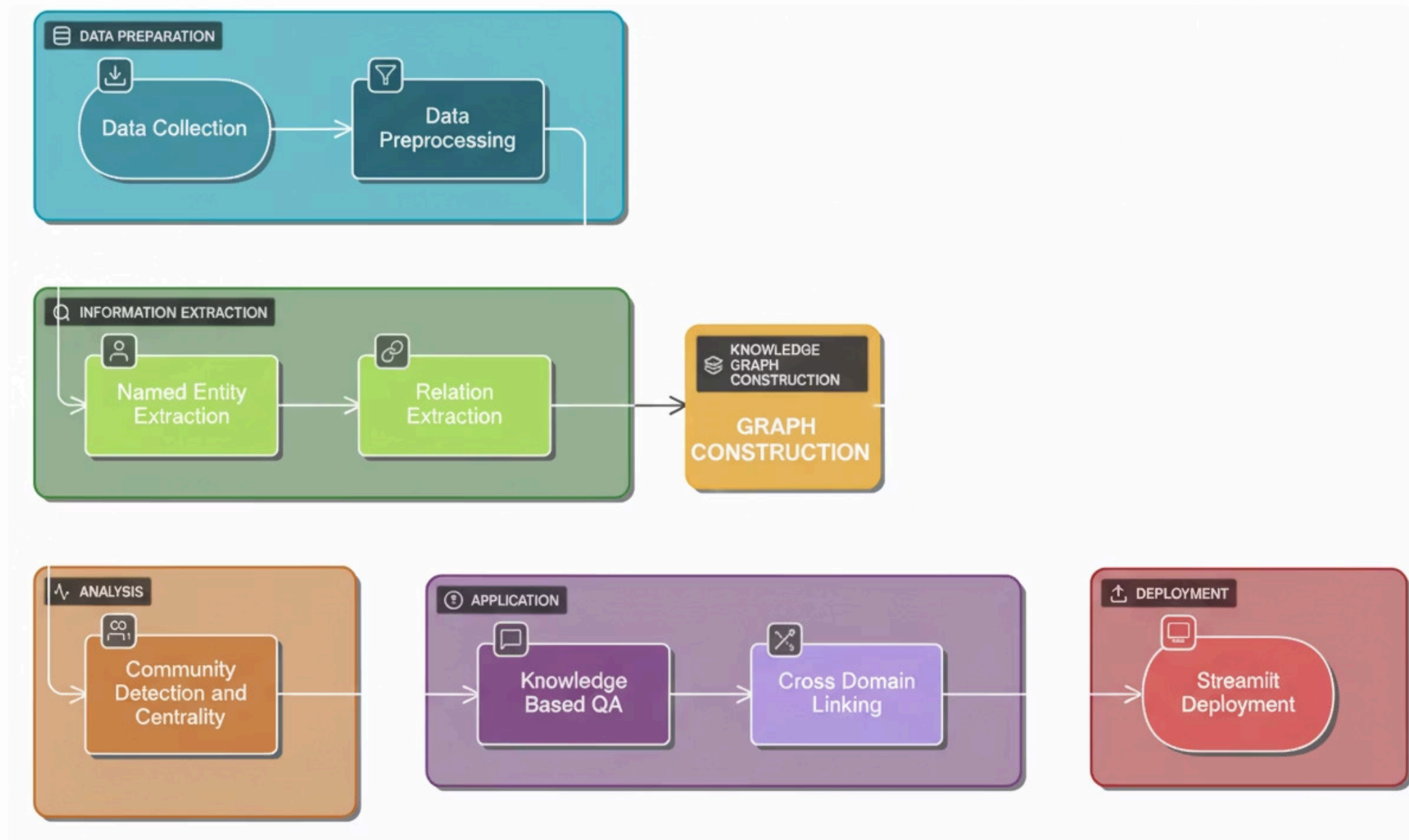
Core Application

KGs transform freeform text into useful, meaningful, and organized information, serving as a compelling tool for solving complex real-world problems.



The Knowledge Graph NLP Workflow

We outline a complete end-to-end workflow, transforming raw text data into a queryable knowledge graph and deploying a practical application.



Step 1: Data Acquisition and Preparation

Gathering Relevant Textual Data

The foundational step involves gathering relevant textual data from diverse sources such as articles, documents, and websites. The quality and diversity of this initial dataset directly impact the richness of the resulting knowledge graph.

Source Diversity

Data can include books, articles, domain-specific documents, or proprietary text.

Focus: Fake News Detection

Our project utilizes the ISOT Fake News Dataset from Kaggle, containing over 25,000 articles.



The dataset includes two types of articles: "True.csv" (12,600+ articles from reputable sources like [reuter.com](https://www.reuters.com)) and "Fake.csv" (12,600+ articles from unreliable sources flagged by Politifact and Wikipedia).

- Each article provides: Title, Text, Type (Real/Fake), and Date.

Steps 2-4: From Text to Structured Graph

After data acquisition, the text must be processed through NLP techniques to distill entities and relationships, culminating in the construction of the graph structure.

2. Entity Recognition

Uses **Named Entity Recognition (NER)** to automatically identify and classify entities (e.g., persons, organizations, locations) within the text.

3. Relation Extraction

Determines the semantic links and relationships between the identified entities. This requires parsing the syntactic and semantic structure of sentences to extract meaningful connections.

4. Graph Construction

The extracted entities become **nodes** and the relationships become **edges**, forming the coherent graph structure (subject–predicate–object triples).

- ❏ While advanced enhancements like integrating **sentiment analysis** or contextual details are possible, they are often complex, time-consuming, and resource-intensive, requiring careful resource allocation.

Graph Analysis: Centrality Measures

In complex networks like knowledge graphs, Centrality and Community Detection uncover hidden structures and identify key influencers, providing deeper insights into knowledge organization.

Centrality: Quantifying Influence

Centrality refers to measures that determine the importance or influence of a node. It answers: *"Which entity plays a central role in connecting different pieces of information?"*



Degree Centrality

Measures the number of direct connections. High degree suggests the entity is a major topic or concept that interacts frequently.



Betweenness Centrality

Measures how often a node lies on the shortest path between others. High betweenness entities act as **critical bridges** or connectors.



Closeness Centrality

Indicates how quickly an entity can reach all other nodes. High closeness suggests rapid influence or access to information across the network.

In this project, centrality highlights the **most influential entities** (key people, events, or sources) in the news graph.

Graph Analysis: Community Detection

Community detection identifies subgroups of nodes that are more densely connected to each other than to the rest of the network, effectively revealing topical clusters.

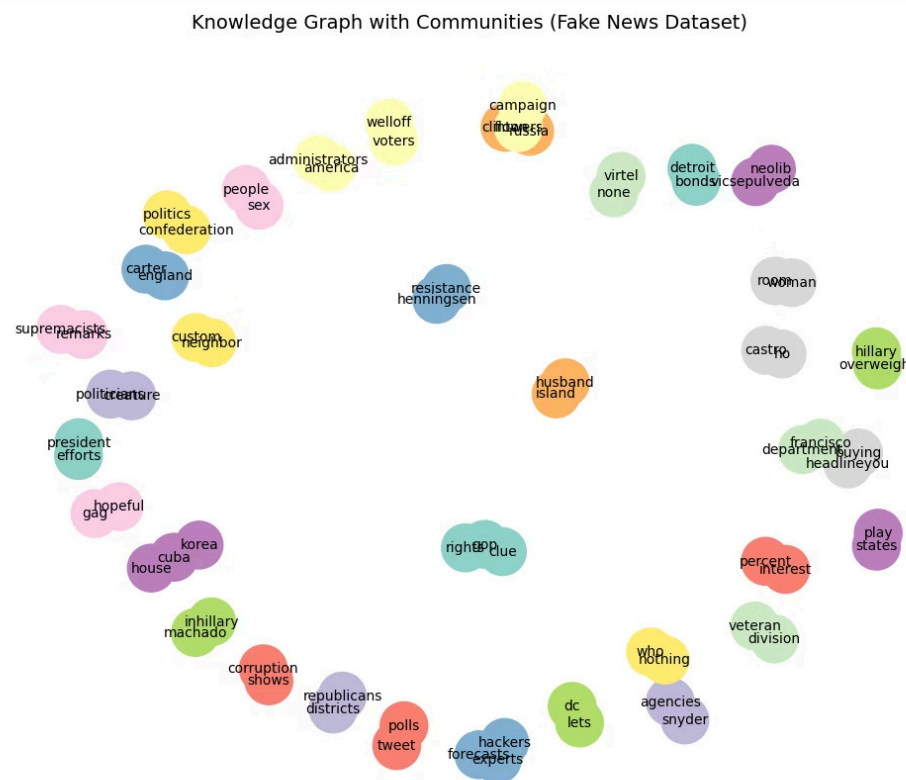
Identifying Topic Clusters

- Nodes within a community share strong, consistent relationships or frequently appear together in the same context.
- Communities help reveal topic clusters, such as groups of entities involved in a specific event or domain.

We used the **Louvain Algorithm**, which detects communities by maximizing **modularity** (dense internal links, sparse external links).

- Automatically assigns each node to a **community ID**.
- Implemented using the `community_louvain` library.

Community detection helps to **group related news topics**—e.g., separating "scientific discoveries" from "political events"—providing a clearer contextual map of the news landscape.



Knowledge-Driven Question Answering (QA)

The QA system enables users to query factual information directly from the constructed knowledge graph, leveraging the structured nature of the data.



User Query

Natural language question is input by the user.



Query Conversion

System processes the query and converts it into a structured graph pattern (Subject, Predicate, Object).



Graph Search

System searches the KG edges (relations) to find matching S–P–O triples.



Result Output

The matched triple or entity is returned as the direct answer.

Factual Representation

The QA system operates on the core unit of the KG: the subject–predicate–object (S–P–O) triple.

Example Triple: (Marie Curie, discovered, Radioactivity)

User Question: “Who discovered radioactivity?”

System Action: Searches for a triple matching (X, discovered, Radioactivity) and returns 'Marie Curie'.

Cross-Domain Linking

The process of connecting related entities and facts across diverse subject areas to create a holistic, contextualized knowledge graph.

What It Means

Connects entities and facts across different knowledge domains (e.g., Politics, Health, Technology) to build a **comprehensive and contextual knowledge graph**.

Strategic Importance

- Exposes **hidden relationships** and misinformation flow, critical for fraud/fake news detection.
- Enhances **semantic understanding** and reasoning capabilities for advanced QA systems.
- Allows analysis to transcend single topics, providing a broader, linked context.

Implementation Workflow



Domain Classification

Tag each article based on its primary subject (Politics, Health, Tech, etc.).



Entity Extraction

Extract all relevant entities and relational triples using NLP tools like SpaCy.



Graph Integration

Merge domain-specific sub-graphs into a unified, **global knowledge graph**.



Multi-Modal Linking

- **Exact Match:** Linking based on identical entity names.
- **Semantic Match:** Linking entities with similar meaning using embeddings (e.g., SentenceTransformer).
- **External Linking:** Aligning entities with external, established sources (Wikidata, DBpedia).

Deployment: Making the Knowledge Accessible

Streamlit Deployment

To make the knowledge graph and its capabilities (like QA and centrality analysis) available to end-users, the application is deployed using Streamlit.



Interactive Dashboard

Provides a simple, elegant web interface for querying the graph and visualizing results.



Rapid Prototyping

Streamlit allows for fast transformation of data scripts into shareable web apps without extensive front-end development.

Accessibility

Enables non-technical stakeholders and researchers to interact with the sophisticated NLP and graph analysis outputs.

Conclusion and Future Scope

Knowledge graphs offer a powerful framework for enhancing NLP by providing semantic context and structure to unstructured text. The presented workflow delivers a functional, queryable system for news analysis.

Key Takeaways

- KGs enable sophisticated reasoning over text data.
- Graph analysis techniques reveal hidden network structures.
- The end-to-end system can be deployed efficiently using modern tools.

Future Directions

- Integrate temporal dynamics into the graph for event tracking.
- Apply Graph Neural Networks (GNNs) for link prediction and entity classification.
- Expand the QA system to handle multi-hop reasoning and complex questions.

Adopting knowledge graph methodology is essential for NLP practitioners seeking to build robust, explainable, and context-aware applications.





Thank You